

# Tendon Transmission

This file computes the tendon-torque mapping matrix  $A$  for the finger and solves for the tendon tensions  $T \geq 0$  that will best produce the reference generalized joint torque  $\tau$ .

## Routing Radii $R$

$R_4$  is the complete routing matrix with 4 rows (splay, MCP, PIP, DIP). Each entry is a pulley radius (mm) denoting the moment arm of a tendon about that joint.

## Generalized PIP/DIP

The angles of PIP and DIP are related to each other by

$$q_{DIP} = kq_{PIP}$$

The generalized torque to  $q_{PIP}$  is then

$$\tau_{PIP_{gen}} = \tau_{PIP} + k\tau_{DIP}$$

And the generalized routing becomes

$$R_{PIP_{gen}} = R_{PIP} + kR_{DIP}$$

## Applying Direction

Routing radii are magnitudes only, so direction (CW vs CCW) is applied by element-wise multiplication to find our signed moment-arm matrix  $A$ .

$$A = D \circ R$$

## Finding Torque from Tension

With  $A$  built, we can determine joint torque via the tendon tension.

$$\tau = AT$$

## NNLS - Solving for Tensions from Reference Torque

Our tendon tensions are constrained by the physical law that they can only pull, not "push", so we impose the constraint

$$T \geq 0$$

We use the solve method Nonnegative Least Squares (NNLS), which the `scipy` Python library provides.

$$\min_{T \geq 0} ||AT - \tau||^2$$

NNLS returns an exact match, when possible, or the closest achievable torque with least-squares.

Optionally, regularization adds a small penalty on tension magnitude, which reduces excessive tensions when there are multiple solutions.

$$\min_{T \geq 0} ||AT - \tau||^2 + \alpha ||T||^2$$